MODERN PHYSICS – MIDTERM 2.

Please fill your **Neptun code** and **Name** on **all papers you submit**. Preferably use these papers for all your calculations as well. If using any extra sheets of paper you submit, write your Neptun code and name on both sides of those too.
Give the answers with a precision of 3 significant digits, together with the units, unless otherwise specified. The score is written for each part of the problem as a sum. Bonus points are marked with a *, they are added to your score, but the total can't exceed 50 points. Some parts of the problems don't rely on the result in earlier parts, make sure to read every part of the problems.

Physical constants

$c = 3.00 \times 10^8\,\mathrm{m/s}$ $\quad h = 6.626\,070\,15 \times 10^{-34}\,\mathrm{J\,s} = 4.135\,667\,7 \times 10^{-15}\,\mathrm{eV\,s}$

$hc = 1.9864 \times 10^{-25}\,\mathrm{J\,m} = 1239.8\,\mathrm{eV\,nm}$ $\quad m_e c^2 = 511\,\mathrm{keV}$

$\lambda = \dfrac{h}{p} = \dfrac{2\pi}{k}$ $\quad \mu_B = e\hbar/2m_\mathrm{e} = 5.788\,38 \times 10^{-5}\,\mathrm{eV/T}$

1. Atomic spin-orbit effect (4 + 6 + 3 + 2* + 3 + 4* = 16 + 6* points)
Rubidium (Rb) has the ground state electron configuration $[\mathrm{Kr}]5\mathrm{s}^1$.

(a) Find the total S, L, J quantum numbers of the ground state of the Rb atom. Label the state using the spectroscopic notation.
Hint: by Hund's 3. rule J is minimized if a subshell is below half-filling, and maximized if it is equal or above.

(b) Two excited states of Rb correspond to the $5s$ electron jumping to the $5p$ orbital, resulting in an electron configuration $[\mathrm{Kr}]5\mathrm{p}^1$. This $5p$ level is split in energy by spin-orbit coupling. What are the total angular momentum quantum numbers j corresponding to the two $5p$ levels? What are the degeneracies of these levels? Label the states using the spectroscopic notation.

(c) Two close spectral lines are observed in the emission spectrum of Rb at wavelengths of $\lambda_1 = 794.8\,\mathrm{nm}$ and $\lambda_2 = 780.0\,\mathrm{nm}$ corresponding to transitions of one electron from the the spin-orbit split $5p$ levels to the ground state $5s$ orbital. Calculate the energy difference between the two $5p$ levels in eV units.

(d) **Bonus:** Draw an energy level diagram showing the spin-orbit splitting, with state labels. Which $5p$ level has the lower energy?

(e) Rb is placed in a weak magnetic field of $B = 0.1\,\mathrm{T}$. Calculate the energy splitting between neighboring levels in all three levels (one $5s$ and the two $5p$). Use the Landé g-factor

$$g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)},$$

and the Hamiltonian for the energy in a magnetic field

$$V = -\vec{\mu} \cdot \vec{B} = g\mu_B B m_j = g\left(\frac{e\hbar}{2m_\mathrm{e}}\right) B m_j.$$

(f) **Bonus:** Draw an energy diagram of the levels showing the (*anomalous Zeeman*) splitting due to the external magnetic field, with state labels. How many spectral lines can be observed?
Hint: recall the selection rules for optical spectra: $\Delta S = 0$, $\Delta L = \pm 1$, $\Delta J = 0, \pm 1$, and $\Delta m_J = 0, \pm 1$, with the additional condition $\Delta m_J \neq 0$ if $m_J = 0$ and $\Delta J = 0$.

2. Dating with ^{14}C (5 points)
In 1947, samples of the Dead Sea Scrolls were analyzed by radiocarbon dating. It was found that the carbon in a sample had an activity of $A = 11\,\mathrm{decays/min}$ coming from the decay of ^{14}C. In contrast, the same amount of carbon from living material exhibits an activity of $A_0 = 14\,\mathrm{decays/min}$. Knowing that the half-life of the ^{14}C isotope is $T_{1/2} = 5.730 \times 10^3\,\mathrm{y}$, how old were the Dead Sea Scrolls in 1947?

3. Scattering and Tunneling (5 + 3 + 2 + 3 = 13 points)
An electron is accelerated to kinetic energy $E = 5\,\mathrm{eV}$, and scatters on a potential barrier of height V_0 and width $L = 4\,\mathrm{nm}$. Outside the barrier, the potential is zero. Let's denote the regions as I, II, and III.

(a) Calculate the de Broglie wavelength λ_I of the electron before it hits the barrier. What is the de Broglie wavelength λ_III on the other side of the barrier? Calculate the corresponding wavenumbers k_I and k_III. Ignore relativistic effects.

(b) Assuming $V_0 = 4\,\mathrm{eV}$, calculate the de Broglie wavelength λ_II, and wavenumber k_II inside the barrier.

(c) Calculate the transmission probability using the equation below. What is the reflection probability?

$$T = \left(1 + \frac{V_0^2 \sin^2(k_\mathrm{II} L)}{4E(E - V_0)}\right)^{-1}.$$

(d) V_0 is increased to $V_0 = 6\,\mathrm{eV}$. Verify the system is in the *wide barrier limit*, and calculate the transmission probability using

$$T = 16\frac{E}{V_0}\left(1 - \frac{E}{V_0}\right)e^{-2\kappa L}, \qquad \kappa = \frac{1}{\hbar}\sqrt{2m_\mathrm{e}(V_0 - E)}.$$

4. Variational Methods: Gaussian potential well (2 + 4 + 4 + 3 + 2 + 1 = 16 points)
Consider an electron in a one-dimensional attractive potential, described by the Hamiltonian

$$H = K + V = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} - V_0\, e^{-\beta x^2},$$

where $\hbar^2/(2m) = 3.810 \times 10^{-2}\,\mathrm{eVnm}^2$, $V_0 = 2\,\mathrm{eV}$ and $\beta = 2\,\mathrm{nm}^{-2}$.
Consider a variational wavefunction, which we choose to also have a Gaussian form, characterized by a single parameter α with nm^{-2} units, and a normalization constant N:

$$\psi(x) = Ne^{-\alpha x^2/2}.$$

(a) Write the normalization condition and show that it is fulfilled for $N = \left(\frac{\alpha}{\pi}\right)^{\frac{1}{4}}$.
Hint: use the integral identity $\int_{-\infty}^{\infty} \mathrm{d}x\, e^{-ax^2} = \sqrt{\frac{\pi}{a}}$.

(b) Calculate the expression for the kinetic energy expectation value $\langle K \rangle_\psi$ that depends parametrically on α.
Hint: use the integral identity $\int_{-\infty}^{\infty} \mathrm{d}x\, x^2 e^{-ax^2} = \frac{\sqrt{\pi}}{2}a^{-\frac{3}{2}}$, *and don't forget the normalization constant.*

(c) Calculate the expression for the potential energy expectation value $\langle V \rangle_\psi$ that depends parametrically on α.

(d) Calculate the expression for the energy expectation value $E(\alpha) = \langle H \rangle_\psi = \langle K \rangle_\psi + \langle V \rangle_\psi$, and apply the variational principle to find an equation for α in the ground state.
Hint: bring the equation to the form $\alpha(\alpha + \beta)^3 = \gamma^2$ *where* $\gamma = 105.0\,\mathrm{nm}^{-4}$ *is a combination of* $\hbar$, m, V_0 *and* β.

(e) This is a complicated equation, but it can be solved in the limit where $\alpha \gg \beta$. Using the approximation $\alpha + \beta \approx \alpha$, find the approximate solution α_0, calculate its numerical value to four significant digits, and verify that $\alpha_0 \gg \beta$.

(f) Calculate the approximate ground state energy $E_0 = E(\alpha_0)$ to four significant digits.
Hint: as a reference value, the numerical solution for this potential yields $E_0 = -1.638\,\mathrm{eV}$.

OFFICIAL SOLUTIONS

1. Atomic spin-orbit effect (4 + 6 + 3 + 2* + 3 + 4* = 16 + 6* points)
Rubidium (Rb) has the ground state electron configuration [Kr]$5s^1$.

(a) Find the total S, L, J quantum numbers of the ground state of the Rb atom. Label the state using the spectroscopic notation.
Hint: by Hund's 3. rule J is minimized if a subshell is below half-filling, and maximized if it is equal or above.

(b) Two close spectral lines are observed at wavelengths of $\lambda_1 = 794.8\,\text{nm}$ and $\lambda_2 = 780.0\,\text{nm}$ corresponding to transitions from the $5s$ to the spin-orbit split $5p$ orbitals. What are the total angular momentum quantum numbers j corresponding to the two $5p$ levels? What is the degeneracy of these levels? Label the states using the spectroscopic notation.

(c) Calculate the energy difference between the two p levels in eV units.

(d) **Bonus:** Draw an energy level diagram showing the spin-orbit splitting, with state labels.

(e) Rb is placed in a weak magnetic field of $B = 0.1\,\text{T}$. Calculate the energy splitting between neighboring levels in all three of the s and the two p levels. Use the Landé g-factor

$$g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)},$$

and the Hamiltonian for the energy in a magnetic field

$$V = -\vec{\mu} \cdot \vec{B} = g\mu_B B m_j = g\left(\frac{e\hbar}{2m_\text{e}}\right) B m_j.$$

(f) **Bonus:** Draw an energy diagram of the levels showing the (*anomalous Zeeman*) splitting due to the external magnetic field, with state labels. How many spectral lines can be observed?
Hint: recall the selection rules for optical spectra: $\Delta S = 0$, $\Delta L = \pm 1$, $\Delta J = 0, \pm 1$, and $\Delta m_j = 0, \pm 1$, with the additional condition $m_j = 0 \nleftrightarrow 0$ if $\Delta J = 0$.

Solution

(a) Rubidium is in an open-shell configuration, with a single spin in the 5s orbital. Therefore, $S = 1/2$ and $L = 0$, and the total orbital momentum takes the value $J = 1/2$. In the spectroscopic notation, the ground state is denoted as $^2S_{1/2}$.

(b) In the first excited state, the 5s electron is excited to the 5p orbital. Therefore, $S = 1/2$ and the lowest-energy state has $L = 1$ (II Hund's rule). The total angular momentum can take values $1/2 = |L-S| \le J \le |L+S| = 3/2$, corresponding to $^2P_{1/2}$ (2-fold degenerate) and $^2P_{3/2}$ (4-fold degenerate).

(c) Since the 5p shell is less-than-half-filled, the state $^2P_{1/2}$ is lower in energy that $^2P_{3/2}$ (III Hund's rule) so that the corresponding transitions (both allowed by the selection rules) have energy given by

$$E = pc = \frac{hc}{\lambda}$$

so that

- $\lambda_1 = 794.8\,\text{nm}$ associated with $^2P_{1/2} \to {}^2S_{1/2}$ ($\Delta m_\ell = -1$ and $\Delta j = 0$) and $E_1 = 1.56\,\text{eV}$
- $\lambda_2 = 780.0\,\text{nm}$ associated with $^2P_{3/2} \to {}^2S_{1/2}$ ($\Delta m_\ell = -1$ and $\Delta j = -1$) and $E_2 = 1.59\,\text{eV}$

Therefore

$$\Delta E = E_2 - E_1 = hc\left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right) = 0.0296\,\text{eV}.$$

(d) The energy diagram showing the spin-orbit splitting and the corresponding transitions is shown in Fig. 1(a).

(e) In the presence of a magnetic field, the energy splitting between any two levels is given by

$$\Delta E = g\mu_B B \Delta m_j$$

where Δm_j is the difference between the total angular momentum quantum numbers of the two states.

Therefore, the (2-fold degenerate) ground state $^2S_{1/2}$ is split into two states with $m_j = -1/2$ and $m_j = +1/2$. The Landé g-factor for this state is $g = 2$ and the splitting is given by $\Delta E = 1.16 \times 10^{-5}$ eV.

Similarly, the (2-fold degenerate) $^2P_{1/2}$ is also split into states with $m_j = -1/2$ and $m_j = +1/2$, but in this case $g = 2/3$ and therefore $\Delta E = 3.86 \times 10^{-6}$ eV, whereas the (4-fold degenerate) $^2P_{3/2}$ is split into states with $m_j = -3/2$, $m_j = -1/2$, $m_j = +1/2$, and $m_j = +1/2$, and with $g = 4/3$ the splitting between states that differ by $\Delta m_J = 1$ is given by $\Delta E = 7.72 \times 10^{-6}$ eV.

(f) There are ten allowed transitions, four between the $^2P_{1/2}$ and $^2S_{1/2}$ states, and six between the $^2P_{3/2}$ $^2S_{1/2}$ states, since the transitions $^2P_{3/2}; m_j = +3/2 \nrightarrow {}^2S_{1/2}; m_j = -1/2$ and $^2P_{3/2}; m_j = -3/2 \nrightarrow {}^2S_{1/2}; m_j = +1/2$, with $\Delta m_j = \pm 2$ are forbidden.
The energy diagram showing the anomalous Zeeman splitting and all the allowed transitions is shown in Fig. 1(b).

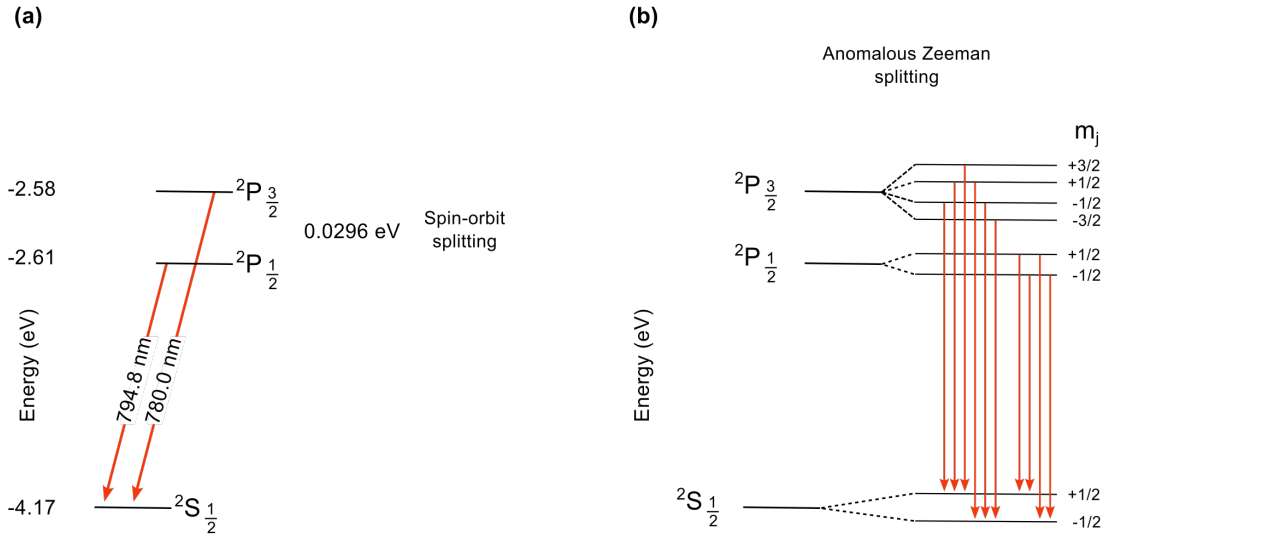


Figure 1: (a) Rb transition lines due to SOI, and (b) in a weak external magnetic field (anomalous Zeeman splitting). Adapted in the style of: `http://hyperphysics.phy-astr.gsu.edu`.

2. Dating with ^{14}C (5 points)
In 1947, samples of the Dead Sea Scrolls were analyzed by carbon dating. It was found that the ^{14}C had an activity of $A = 11\,\text{decay/min}$. In contrast, living material exhibits an activity of $A_0 = 14\,\text{decay/min}$. Knowing that the half-life of the ^{14}C isotope is $T_{1/2} = 5.730 \times 10^3\,\text{y}$, how old are the Dead Sea Scrolls?

Solution
Radioactivity follows the decay law

$$N = N_0 e^{-\lambda t} = N_0 e^{-\ln(2) t/T_{1/2}}$$

where, in the case of ^{14}C, the decay constant is given by $\lambda = \ln(2)/T_{1/2} = 1.21 \times 10^{-4}\,\text{y}^{-1}$.

The activity is defined as

$$A = -\frac{dN}{dt} = \lambda N.$$

The ratio of the activities then gives

$$\frac{A}{A_0} = \frac{N}{N_0} = e^{-\lambda t}$$

so that

$$t = -\ln\left(\frac{A}{A_0}\right)\frac{1}{\lambda} = 1.99 \times 10^3\,\text{y}.$$

From the measurement performed in 1947, the Dead Sea Scrolls were determined to be 2000 years old, giving them a date of 53 BC, and confirming their authenticity.

3. Scattering and Tunneling (5 + 3 + 2 + 3 = 13 points)
An electron is accelerated to kinetic energy $E = 5\,\mathrm{eV}$, and scatters on a potential barrier of height V_0 and width $L = 4\,\mathrm{nm}$. Outside the barrier, the potential is zero. Let's denote the regions as I, II, and III.

(a) Calculate the de Broglie wavelength λ_I of the electron before it hits the barrier. What is the de Broglie wavelength λ_III on the other side of the barrier? Calculate the corresponding wavenumbers k_I and k_III. Ignore relativistic effects.

(b) Assuming $V_0 = 4\,\mathrm{eV}$, calculate the de Broglie wavelength λ_II, and wavenumber k_II inside the barrier.

(c) Calculate the transmission probability using the equation below. What is the reflection probability?

$$T = \left(1 + \frac{V_0^2 \sin^2(k_\mathrm{II} L)}{4E(E - V_0)}\right)^{-1}.$$

(d) V_0 is increased to $V_0 = 6\,\mathrm{eV}$. Verify the system is in the *wide barrier limit*, and calculate the transmission probability using

$$T = 16\frac{E}{V_0}\left(1 - \frac{E}{V_0}\right)e^{-2\kappa L}, \qquad \kappa = \frac{1}{\hbar}\sqrt{2m_\mathrm{e}(V_0 - E)}.$$

Solution
The problem is governed by the time-independent Schrödinger equation

$$\frac{d^2\psi(x)}{dx^2} = -\frac{2m_\mathrm{e}}{\hbar^2}\left[E - V(x)\right]\psi(x) = \beta^2\psi(x),$$

with

$$\beta^2 = \frac{2m_\mathrm{e}}{\hbar^2}\left[V(x) - E\right].$$

(a) In regions I and III, $V(x) = 0$ and $\beta^2 = -\frac{2m_\mathrm{e}}{\hbar^2}E < 0$ so that $\beta \in \mathbb{C}$ is a *complex* number the Schrödinger equation admits an oscillating solution for $\beta = \pm ik$ of the form

$$\psi_\mathrm{I,III}(x) \propto e^{ik_\mathrm{I,III}\,x}$$

with wavenumber is the same in the two regions because in both, the potential is zero

$$k_\mathrm{I} = k_\mathrm{III} = \frac{1}{\hbar}\sqrt{2m_\mathrm{e}E} = \frac{2\pi}{hc}\sqrt{2m_\mathrm{e}c^2E} = 11.35\,\mathrm{nm}^{-1}$$

and the corresponding de Broglie wavelength is

$$\lambda_\mathrm{I} = \frac{2\pi}{k_\mathrm{I}} = \lambda_\mathrm{III} = \frac{2\pi}{k_\mathrm{III}} = 0.548\,\mathrm{nm}$$

(b) Note that in this case, in region II, $E > V_0$ so that it remains true that $\beta \in \mathbb{C}$ but the wavenumber decreases

$$k_\mathrm{II} = \frac{1}{\hbar}\sqrt{2m_\mathrm{e}(E - V_0)} = \frac{2\pi}{hc}\sqrt{2m_\mathrm{e}c^2(E - V_0)} = 5.12\,\mathrm{nm}^{-1}$$

and the corresponding de Broglie wavelength increases

$$\lambda_\mathrm{II} = \frac{2\pi}{k_\mathrm{II}} = 1.23\,\mathrm{nm}$$

(c) Calculating the transmission probability yields $T = 0.556 < 1$ and, due to charge conservation, the reflection probability is given by $R = 1 - T = 0.444 > 0$. This is in stark contrast with the classical case, in which one expects $T_\mathrm{classical} = 1$ and $R_\mathrm{classical} = 1 - T_\mathrm{classical} = 0$.

(d) Once the potential increases in region II, $E < V_0$ so that $\beta \in \mathbb{R}$ and the Schrödinger equation admits an exponentially suppressed solution of the form

$$\psi_\mathrm{II}(x) \propto e^{-k_\mathrm{II}\,x}.$$

The problem is in the *wide barrier limit* as

$$\kappa L = 20.5 \gg 1.$$

It follows that the transmission probability is $T = 3.47 \times 10^{-18} > 0$.

4. Variational Methods: Gaussian potential well (2 + 4 + 4 + 3 + 2 + 1 = 16 points)
Consider an electron in a one-dimensional attractive potential, described by the Hamiltonian

$$H = K + V = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} - V_0\, e^{-\beta x^2},$$

where $\hbar^2/(2m) = 3.810 \times 10^{-2}\,\mathrm{eVnm}^2$, $V_0 = 2\,\mathrm{eV}$ and $\beta = 2\,\mathrm{nm}^{-2}$.
Consider a variational wavefunction, which we choose to also have a Gaussian form, characterized by a single parameter α with nm^{-2} units, and a normalization constant N:

$$\psi(r) = Ne^{-\alpha x^2/2}.$$

(a) Write the normalization condition and show that it is fulfilled for $N = \left(\frac{\alpha}{\pi}\right)^{\frac{1}{4}}$.
Hint: use the integral identity $\int_{-\infty}^{\infty} \mathrm{d}x\, e^{-ax^2} = \sqrt{\frac{\pi}{a}}$.

(b) Calculate the expression for the kinetic energy expectation value $\langle K\rangle_\psi$ that depends parametrically on α.
Hint: use the integral identity $\int_{-\infty}^{\infty} \mathrm{d}x\, x^2 e^{-ax^2} = \frac{\sqrt{\pi}}{2}a^{-\frac{3}{2}}$, *and don't forget the normalization constant.*

(c) Calculate the expression for the potential energy expectation value $\langle V\rangle_\psi$ that depends parametrically on α.

(d) Calculate the expression for the energy expectation value $E(\alpha) = \langle H\rangle_\psi = \langle K\rangle_\psi + \langle V\rangle_\psi$, and apply the variational principle to find an equation for α in the ground state.
Hint: bring the equation to the form $\alpha(\alpha+\beta)^3 = \gamma^2$ *where* $\gamma = 105.0\,\mathrm{nm}^{-4}$ *is a combination of* $\hbar$, m, V_0 *and* β.

(e) This is a complicated equation, but it can be solved in the limit where $\alpha \gg \beta$. Using the approximation $\alpha + \beta \approx \alpha$, find the approximate solution α_0, calculate its numerical value, and verify that $\alpha_0 \gg \beta$.

(f) Calculate the approximate ground state energy $E_0 = E(\alpha_0)$.
Hint: as a reference value, the numerical solution for this potential yields $E_0 = -1.638\,\mathrm{eV}$.

Solution

(a) The normalization condition reads

$$1 = \int_{-\infty}^{\infty} \mathrm{d}x\, \psi^*(x)\psi(x) = N^2 \int_{-\infty}^{\infty} \mathrm{d}x\, e^{-\alpha x^2} = N^2 \left(\frac{\pi}{\alpha}\right)^{1/2}$$

thus yielding

$$N = \left(\frac{\alpha}{\pi}\right)^{\frac{1}{4}}$$

(b) The expectation value of the kinetic energy is given by

$$\langle\psi|\, K\, |\psi\rangle = -\frac{\hbar^2}{2m}\left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} \mathrm{d}x\, e^{-\alpha x^2/2}\left(\frac{\partial^2}{\partial x^2} e^{-\alpha x^2/2}\right) = -\frac{\hbar^2}{2m}\left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} \mathrm{d}x\, e^{-\alpha x^2}\left[-\alpha\left(1-\alpha x^2\right)\right] = \frac{\hbar^2}{4m}\alpha$$

(c) The expectation value of the potential energy is given by

$$\langle\psi|\, V\, |\psi\rangle = -V_0\left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} \mathrm{d}x\, e^{-(\alpha+\beta)x^2} = -V_0\frac{\alpha^{1/2}}{(\alpha+\beta)^{1/2}}$$

(d) The parametric expression for ground state energy reads

$$E(\alpha) = \frac{\hbar^2}{4m}\alpha - V_0\sqrt{\frac{\alpha}{(\alpha+\beta)}},$$

for which one can verify that each term has the proper physical dimensions.
The variational principle implies that

$$\frac{d}{d\alpha}E(\alpha) = \frac{\hbar^2}{4m} - V_0\frac{1}{2}\left[\frac{1}{\alpha^{1/2}(\alpha+\beta)^{1/2}} - \frac{\alpha^{1/2}}{(\alpha+\beta)^{3/2}}\right] = 0$$

By rearranging the terms and squaring both the left- and right-hand sides one gets

$$\left(\frac{\hbar^2}{2mV_0}\right)^2 = \left[\frac{1}{\alpha(\alpha+\beta)} + \frac{\alpha}{(\alpha+\beta)^3} - \frac{2}{(\alpha+\beta)^2}\right] = \frac{\beta}{\alpha(\alpha+\beta)^3}.$$

Therefore, the equation reduces to

$$\alpha(\alpha+\beta)^3 = \gamma^2,$$

where

$$\gamma^2 = \beta^2 \left(\frac{2mV_0}{\hbar^2}\right)^2.$$

(e) In the limit $\alpha \gg \beta$ the expression above reduces to

$$\alpha^4 = \gamma^2$$

which admits the solution

$$\alpha_0 = \sqrt{\gamma} = 10.25\,\mathrm{nm}^{-2} \gg \beta.$$

(f) Substituting the optimal value $\alpha = \alpha_0$ in the parametric expression for the ground state energy, one finds

$$E_0 = E(\alpha_0) = -1.634\,\mathrm{eV}$$

which differs from the numerical value by less than 1 %.